Radar Tracking Software Pipeline Optimisation for UAV Detect-and-Avoid

NOAH WEST

B.Eng (Hons)

Supervisor: Dr. Donald Dansereau

A thesis submitted in fulfilment of
the requirements for the degree of
Bachelor of Engineering (Honours)

School of Aerospace Mechanical and Mechatronics Engineering
Faculty of Engineering
The University of Sydney
Australia

16 March 2026

Abstract

- Rodar is able to provide Unmanned Aeriel Vehicales with the capability to detect and track airborne hazzards over large distances and in all weather conditions. Talk about trade off between size, power, weight and compute. Talk about problems.
- We seek to address these problems through adapting establish radar processing algorithms for UAVs.
- By evaluating the effectiveness of different implementations and algorithms essential to multi target tracking, this report demonstrates how software choises make long range radar practical for small UAV detect and avoid systems.
- Talk about key parts of our pipeline
- Talk about the Experiments
- Talk about findings
- Ultimately, our work provides a reference for engineers and policy makers...

Statement of Contributions

I, **Noah West**, declare that the intellectual content of this thesis is the product of my own work and that all assistance received in the preparation of this thesis and relevant sources have been appropriately acknowledged. The contributions I made to this research include:

- Undertaking the background research and completing the literature review.
- Defining the research scope and developing the problem statement in consultation with my supervisor and the industry partner.
- Designing the methodology, experimental approach, and test plan with guidance from my supervisor.
- Designing and implementing, in collaboration with engineers from the industry partner *Mission Systems*, the software pipeline described in this work.
- Profiling and evaluating the pipeline software to produce the reported results.
- Capturing radar data for use in testing and validation.
- Conducting the experiments, including configuration of the test setup and execution of data collection procedures.
- Performing analysis of results, including quantitative evaluation using appropriate metrics to validate conclusions.
- Interpreting findings, and writing the discussion and conclusions, incorporating feedback from my supervisor and relevant stakeholders.
- Writing the thesis and preparing figures, tables, and supporting material.

The above represents an accurate summary of my contribution.

Noah West
Honours Candidate
School of Aerospace, Mechanical and Mechatronic Engineering,
The University of Sydney
Date: 16th March 2026

Dr. Donald Dansereau
Senior Lecturer
Australian Centre for Field Robotics,
The University of Sydney
Date: . . . . . .

Acknowledgements

I would like to express my deepest gratitude to my supervisor, Dr. Donald Dansereau, for his guidance and support throughout this research.

I am also thankful to my industry mentors Rex, Hugh, and David Johnson. This work would not be possible without their enthusiastic support, and expertise.

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CHAPTER 1

Introduction

1.1 Motivation and Context

With more uncrewed aerial vehicles (UAVs) in the air each year, and a growing desire for them to operate in controlled airspaces, there is a concern regarding their ability to maintain separation from hazards in the air. United States and European federal agencies are developing policy that necessitates detect-and-avoid (DAA) systems for UAVs, which parallel the visual and instrument flight rules (VFR & IFR) that dictate how human pilots avoid traffic [1] [2] [3].

While some proposed DAA systems utilise optical and infrared sensors [3], radar sensors may also be suitable, as they can perform sensing over great distances in all weather conditions, due to their relatively low atmospheric attenuation (compared with optical systems) [4]. The inconvenient side effect of their long wavelength property is that angular resolution is much lower compared to similarly sized optical systems. If this challenge is overcome, the implementation of radar as the primary sensing mode or an additional sensing mode will improve the situational awareness, and thus the safety, of UAVs.

The main historical objection towards radar has not been with its usefulness, but with its fit within the SWaP constraints of smaller aircraft. This historical constraint is now weakening because both sides of the radar stack have miniaturised. Texas Instrument's AWR2243 is a single 76-81 GHz radar chip that integrates phase-locked loops (PLLs), three transmitters, four receivers, a mixer, an ADC, and self calibration, all on a 10.4mm

10.4mm printed circuit board (PCB)-friendly package. Higher frequency, lower noise, electronics and the miniaturisation of the transistor have been key enablers of this technology. Higher frequency radar allows of higher resolution imaging or smaller aperture; lower noise allows for an increase sensitivity to targets. On the compute side, modern UAV systems often have a dedicated, high compute companion computer in addition to the flight compute. NVIDIA's Jetson series computers such as Xavier NX, Orin Nano and Orin NX providing up 21, 67, 157 trillion operations per second (TOPS) on 8-bit integers respectively, while each weighting under 50g (without a heat sink).

Plextek's mmWave-based 60 GHz sense-and-avoid radar for UAVs demonstrates the feasibility of pairing miniaturised radar hardware with modern edge computing for the detection of people, vehicles, poles, and buildings at ranges of hundreds of metres. The wider UAV market shows a similar shift toward more computationally demanding onboard tasks. Skydio's X10, for instance, incorporates both an NVIDIA Jetson Orin SoC and a Qualcomm QRB5165 SoC within a 2.11 kg platform while retaining up to 40 minutes of flight time, illustrating that substantial embedded compute is now compatible with practical UAV SWaP limits. These developments suggest that recent improvements in radar miniaturisation and embedded heterogeneous computing are promoting a new class of computation-heavy UAV sensing systems, making it timely to revisit radar target tracking and to determine the most suitable signal-processing pipeline for UAV detect-and-avoid applications.

Despite the challenges, UAVs are the fastest growing market for radar systems, projected to grow with a compound annual growth rate (CAGR) of 11.22% globally through 2030 [5]. To make these emerging radar systems more performant, accessible, and safe, it would be valuable to find the most suitable radar signal processing pipeline for UAV DAA systems, revisiting the old problem of radar target tracking in the face of unique hardware conditions.

Software-defined architecture (SDAs) are increasingly preferred over hardware defined radar systems for UAV applications as software modifications can be shipped faster than hardware modifications [5]. As a result, graphics processing units (GPUs) are favoured over non-programmable digital signal processors such as application specific integrated circuits (ASICs). GPU-accelerated implementations are also favourable for computationally intensive stages of

the radar processing chain, as many core operations, such as fast Fourier transforms (FFTs) and matrix operations, are highly data-parallel, allowing for large increases in throughput when compared to serial processors. Additionally, GPU-based pipelines can leverage mature developer friendly, GPU software ecosystems and optimised libraries [6]. Additionally, the incorporation of AI into radar tracking algorithms has further encouraged these highly parallel, GPU-based pipelines [7]. Key open questions remain around balancing the compute saved through GPU parallelism against the cost of data movement and synchronisation, and how unified (shared) CPU–GPU memory on embedded platforms changes this trade-off.

Given the miniaturisation of radar enabling technology, the potential benefits for radar base DAA systems, and the range limitations of systems suitable for small radar systems. This report proposes a software pipeline design that allows for the radar tracking of aerial hazards on UAVs with strict SWaP constraints. Particular emphasis will be placed on maximising the sensitivity to faraway targets, a requirement for operating in shared airspace, and minimising computation resources and latency. The knowledge gaps that will be addressed are detailed in the following section.

1.2 Knowledge Gap

TODO: This section is to be written after the Literature review is finished

TODO: Talk broadly about the state of the current literature and how they engage with this new issue. Summarise the major gaps in the literature in terms of radar systems under swap constraints, and if the literature has missed any techniques that we're going to employ. Tie that into our aims and outcomes, explaining how our experiments will fill the gaps.

1.3 Problem Definition

Small-UAV radar systems often face two coupled technical difficulties when compared to larger radar systems: reduced signal-to-noise ratio (SNR), which weakens sensitivity to small or distant targets, and reduced angular resolution, which makes it harder to separate, localise, and confidently track targets in cluttered or multi-target scenarios. While recent technological developments in compact radar hardware and light edge computing have made such radar systems feasible, it is clear from the discussion above that there is no clear systems level consensus for the most suitable software system design for the unique SWaP constraints of small UAVs.

This thesis addresses that gap by defining the problem as one of *hardware-aware radar pipeline design for UAV detect-and-avoid*. Specifically, the problem is to determine how a software-defined radar tracking pipeline for a small monopulse radar should be structured and implemented so that it can reliably detect and track faint aerial targets on UAV-compatible hardware. This includes identifying suitable combinations of front-end signal processing, detection logic, angle estimation, data association, and tracking filters, while also accounting for the computational effects of GPU acceleration, host–device data movement, synchronisation overhead, and differing embedded memory architectures. In this thesis, success is defined by the joint ability of a pipeline to maintain target sensitivity and track quality while meeting practical embedded performance requirements.

1.4 Scope and Assumptions

TODO: Trim down this sections. It's way too verbose

This thesis is concerned with the *software design and evaluation* of radar tracking pipelines for UAV detect-and-avoid under size, weight and power constraints. Its scope is limited to the processing stages that occur after radar samples have been acquired, with emphasis on how alternative signal-processing, angle-estimation, and tracking algorithms perform when

implemented on embedded compute platforms suitable for small and medium UAVs. The work therefore focuses on hardware-aware software design rather than radar hardware design itself. In particular, the thesis evaluates candidate pipelines through three main forms of analysis: profiling of individual algorithms, profiling of integrated end-to-end pipelines, and sensitivity analysis of detection and tracking performance under constrained signal conditions. These evaluations are used to compare cost and performance in terms of latency, throughput, memory behaviour, and sensitivity to faint or distant targets.

Within this scope, the processing chain considered includes front-end shaping and windowing, fast Fourier transforms, coherent integration, constant false alarm rate detection, monopulse angle sensing, coordinate transformation, data association, tracking filters, and track management. The thesis also considers orchestration and dataflow decisions, such as batching, reading and writing strategies, stream aggregation, and the performance implications of host–device transfers and synchronisation. These elements are included because the identified knowledge gap is not only algorithmic, but systems-level: the suitability of a radar tracking pipeline for UAV detect-and-avoid depends jointly on sensing performance and implementation cost on embedded hardware.

TODO: This is a bit repetative

The empirical scope includes both controlled and partially realistic evaluation. Candidate methods are tested using a combination of simulation, synthetic or simplified input data, and captured radar data where available, in order to study both low-level computational behaviour and tracking performance in encounter-relevant scenarios. Partial field testing is included only to the extent necessary to demonstrate practical feasibility and to expose the pipeline to non-ideal data. The thesis does not attempt a complete flight-qualified validation campaign, nor does it claim certification readiness for operational UAV detect-and-avoid deployment. Rather, the aim is to provide comparative evidence and design guidance for practical onboard radar processing under realistic embedded constraints.

TODO: Idk if I need the word "explicitly"

Several topics are explicitly outside the scope of this work. First, the thesis does not address the design of the radar front-end hardware, including antenna design, RF electronics, waveform generation, or board-level integration. Second, it does not provide full-scale field trials across all weather, geometry, clutter, and airspace conditions that would be required for operational certification. Third, although FPGA- and ASIC-based implementations are relevant to radar systems, they are not the focus here; the work is restricted to programmable CPU/GPU-oriented platforms in order to study software-defined architectures that can be iterated rapidly and deployed on modern embedded UAV compute stacks. Finally, this thesis does not attempt a cross-modality comparison between radar and optical or infrared sensing. Optical and infrared systems are discussed only to motivate radar's complementary role, not to benchmark relative sensing performance across modalities.

The thesis proceeds under several assumptions. The first is that the radar sensor is a *small conformal monopulse radar* whose aperture and power limitations are representative of a UAV-compatible installation. This assumption is consistent with the thesis focus on low-SNR operation and reduced angular resolution relative to larger airborne radar systems. The second is that the target compute platform includes *GPU-enabled embedded hardware*, reflecting the increasing use of heterogeneous CPU/GPU companion computers in modern UAV systems and the thesis emphasis on highly parallel radar processing. The third is that the hardware context of primary interest includes *shared or unified memory* between CPU and GPU, as found in many embedded systems-on-module, although comparison against CPU-only and discrete-memory contexts may be used where available to clarify architectural trade-offs. These assumptions are adopted because they reflect the practical hardware class for which the pipeline is being designed: not a large airborne radar platform with abundant power and aperture, but a compact UAV-compatible software-defined radar system in which memory movement, synchronisation overhead, and implementation structure materially affect performance.

Under these boundaries, the thesis does not seek to produce a universally optimal radar architecture. Instead, it seeks to identify which combinations of algorithms and implementation strategies are most suitable for the stated UAV context, and how those conclusions vary with embedded hardware restrictions. The outcome is intended to be a set of hardware-aware recommendations for the design of practical radar tracking pipelines for UAV detect-and-avoid, rather than a complete end-to-end certified system design.

1.5 Contributions

TODO: Write aims and outcomes section heading/intro.

The proposed thesis will be a systems report regarding the software design space of a GPU-based radar tracking pipeline for a small monopulse radar. The problems of poor resolution and low signal strength described above will be addressed in the proposed project, a systems report that compares different signal processing and tracking algorithms for their ability to detect faint targets from small UAV platforms. Due to the growing interest in the hardware for UAV applications, these algorithms will be implemented on GPUs and their throughput and latency will be measured. Questions about the cost of memory movement and synchronisation will be addressed through integration testing and profiling of the entire pipeline. A portion of the research will also contribute to an understanding of the effect that different hardware contexts on the optimal design of these radar processing pipelines.

In summary, the project will contribute the following:

- A comparison of candidate detection, angle-estimation, and tracking algorithm combinations, assessed by detection sensitivity and track quality in multi-target aerial scenarios, to inform algorithm selection.
- A comparison of alternative low-level implementations of the selected algorithms, quantifying their effects on end-to-end throughput and latency in highly parallelised GPU pipelines, to inform implementation choices.

- An assessment of how these algorithm and implementation conclusions vary across embedded hardware configurations, providing hardware-aware design recommendations.

TODO: Add any missing aims/outcomes content that is not present in the provided text.

TODO: Tie the aims and outcomes into the motivation

1.6 Thesis Structure

Chapter 2 introduces the background information related to the following areas: standard radar algorithms, monopulse angle estimation, multi-target tracking, and parallel processing.

Chapter 3 conducts and extensive literature review of both foundational and state of the art approaches to performant radar systems under high SWaP constraints. The methodologies of works with similar motivations to ours are critically evaluated for their effectiveness in meeting their aims. Key findings and gaps across works are summarised and an explanation for how this thesis fits in with the existing literature is provided.

Chapter 4 Explains the methodology by which we compared different radar processing pipeline designs for their suitability for use in detect and avoid systems, and compare different algorithms for their ability to detect small, faraway targets. This methodology will describe the algorithms, the implementations of these algorithms, the processes by which the implementations are profiled, and the processes by which the algorithms are compared for their impact on the overall sensitivity of the system.

Chapter 5 Presents the results associated with each of the experiments outlined in the methodology. The results are illustrated as tables and figures, with descriptions of the findings of each of the methodology's stages.

Chapter 6 The key findings in the results are discussed. The results of different experiment stages are evaluated and explained. Recommendations are provided. The results are tied in with the problem and contributions.

Chapter 7 The key findings are summarised and the experiment is examined for any areas for improvement or future work. The value of the work is assessed for its contributions to the industry partner, and wider contributions to the field of UAV DAA.

TODO: Do I include appendix content like WHS report?

CHAPTER 2

Background

2.1 Aviation Policy

- Part 91 of the Civil Aviation Safety Regulations (CASR) from the Civil Aviation Safety Authority (CASA) of Australia sets out the general operating rules for the pilots and operators of all aircraft, with the key exception of UAVs, which are regulated separately under Part 101. While Part 91 outlines a class-based framework that affords different levels of freedom of operation based on the type of manned aircraft and the type of airspace, Part 101 imposes a narrrower set of permissions, limiting the maximum flying height to 120 meters, only within aerodromes, vastly limiting their operations. Manned aircraft fall under two sets of rules: Visual Flight Rules (VFR), which come into effect when navigation is performed visually, and Instrument Flight Rules (IFR), where a pilot is flying primarily by reference to instruments. These rules are permissive in that they allow for flight in certain classes of airspace without special permission. There is no such framework for UAVs, regardless of how robust their navigation and DAA systems are.
- The UAV Traffic Management (UTM) Plan point towards incremental reform that so that drones and new technology can utilise airspace safely and productively. The plan stresses the importance autonomous onboard DAA systems in maintaining a safe airspace. Each UAV should have the capability to sense both cooperative, Automatic Dependent Surveillance-Broadcast (ADS-B) equipped aircraft and non-cooperative aircraft. The plan involves risk-based action by the regulator, with the aim of promoting a reliable, efficient and safe ecosystem.

- In order to facilitate the introduction of UAVs into controlled airspace under this plan, a proven DAA system will be required to meet the standards of VFR or IFR flight.
- Through the design of a reliable and small radar system, this thesis aims to contribute to the development of DAA systems that can meet the standards of VFR or IFR flight, and thus facilitate the safe integration of UAVs into controlled airspace.

TODO: Question: Should I discuss the state of policy across, USA, Canada, Europe. Which contries share policy and which ones are leading the adoption of UAVs.

TODO: Discuss the current requirements for DAA systems. Describe what it would take for them to meet the standards of instrument or visual flight rules.

TODO: Is this the most appropriate place for policy to be discussed? It's important for motivation and goals but it isn't relevant to understanding the technology.

2.2 Monopulse Radar Hardware

- Radar is a active sensing system that utilises radio waves to determine the distance, direction and velocity of objects in the environment.
- Monopulse radar is a specific radar technique that, allows for the estimation of target position from a single transmitted pulse (as the name suggests). This is achieved by comparing the received signal across multiple receiver antennae simultaneously, rather than sequentially as seen in mechanical scanning radar systems. As a results, monopulse radar can scan targets rapidly and with low latency.
- Another quality of monopulse radar is that it can be implemented using phased array antennas, which allow for electronic beam steering without moving parts.

- While the electronics are complex, the mechanical simplicity of a solid-state radar system can be advantageous for UAV applications, where size, weight and power constraints are critical. This combined with the low latency tracking makes monopulse radar suitable for DAA systems
- This single-look angle measurement capability is valuable for DAA applications, where encounter geometries can change quickly and the sensing system must provide timely and stable bearing information. (TIMING DIAGRAM MAYBE)
- Radar systems are fundamentally limited by the diffraction of radio waves, which sets a lower bound on the angular resolution that can be achieved for a given antenna aperture and operating frequency.
- The angular resolution limit is often expressed as the beamwidth, which is approximately proportional to the wavelength divided by the antenna diameter. This means that for a given frequency, a larger antenna can achieve finer angular resolution. Due to the infeasibility of large antennas on small UAVs, we need to have an angular accuracy that is higher than the beamwidth, which is possible with angle discrimination, a core principle of monopulse radar.
- Figure (MONOPULSE FIG) illustrates how monopulse radar can achieve angle estimation accuracy that is finer than the beamwidth by comparing the received signal across multiple channels. (INCLUDE FIGURE)
- Monopulse processing may be implemented in amplitude-comparison form, phase-comparison form, or hybrid forms depending on the antenna architecture, frequency band, and required accuracy.
- The achievable angular accuracy of a monopulse radar depends on several factors, including antenna aperture, frequency, signal-to-noise ratio, and calibration quality.
- In general, narrower beams and higher signal-to-noise ratios lead to better angle estimation accuracy because the monopulse slope near boresight becomes steeper and measurement noise has less influence.
- Important error sources include thermal noise, receiver imbalance, gain and phase mismatch between channels, quantisation effects, multipath, clutter, sidelobe interference, and target glint.

- Simulation is therefore important for characterising expected performance before hardware deployment.
- The performance of digital processing stages such as amplitude comparison will be evaluated in terms of its resulting angular accuracy, range accuracy, and ability to track multiple targets, under simulation and real world conditions.

2.3 Radar Processing

- Radar processing converts received echoes into target measurements such as range, velocity, and angle.
- Radar hardware performs transmission, reception, amplification, pulse-compression, downconversion, and analogue-to-digital conversion, while software extracts useful target information from sampled data.
- In software, a typical processing chain begins by organising the pulse-compressed samples across fast time, slow time, and antenna channels.
- By considering the samples the spectral domain (either with matched filtering or a fast time Fourier Transform), we can estimate the range of targets.
- Optionally, a long time Fourier transform can be applied across slow time to estimate Doppler, which provides target radial velocity and helps separate moving targets from clutter.
- Coherent integration across multiple pulses improves detectability of weak targets. This involves summing the complex samples across pulses.
- After integration detection algorithms such as Constant False Alarm Rate (CFAR) can identify candidate target returns, which are then passed to angle estimation and tracking stages.
- The bearing and elevation of targets can be estimated using monopulse processing, which compares the received signal across multiple antenna channels to infer angle with high accuracy.
- The resulting range, Doppler, and angle measurements form the basis for later tracking and classification stages in a DAA system.

$$f(x) = \frac{3}{2}$$

2.4 Multi Target Tracking

- Multi-target tracking converts a sequence of noisy radar detections into stable target tracks over time.
- Its purpose is to estimate the current state of each target, including position and velocity, while maintaining continuity between successive radar updates.
- In radar systems, tracking is typically performed after detection and angle estimation, using range, Doppler, and angle measurements as inputs.
- A tracking filter is used to smooth noisy measurements and predict target motion between radar updates.
- Kalman filtering provides a common framework for this process by combining a motion model with uncertain measurements to produce an improved estimate of target state.
- The filter alternates between prediction and update stages, using the predicted target state to guide association and the new measurements to refine the estimate.

- Variants such as the Extended Kalman Filter may be required when the measurement model is nonlinear.
- Radar measurements are naturally produced in polar coordinates, typically as range, bearing, elevation, and sometimes Doppler.
- For tracking, these measurements may be processed either directly in polar form or transformed into Cartesian coordinates depending on the filter design and motion model.
- The choice of coordinate system affects the linearity of the model, the form of the measurement uncertainty, and the complexity of the update equations.
- Gating is used to restrict which detections are considered plausible matches for an existing track.
- This reduces computational cost and helps reject unlikely associations caused by clutter or nearby targets.
- Data association then determines which measurement should be assigned to each predicted track when multiple candidates are present.
- Track life management controls how tracks are initiated, confirmed, maintained, and deleted over time.
- New detections may first form tentative tracks, which are only promoted to confirmed tracks after consistent observations across multiple updates.
- Similarly, tracks may be deleted after repeated missed detections to avoid maintaining false or stale targets.
- These stages allow the tracker to balance responsiveness to new targets with robustness against clutter and intermittent measurement loss.

2.5 Parallel Processing

- Parallel processing is important in radar systems because many stages of the pipeline, such as FFTs, beamforming, matrix operations, and detection, involve applying the same operation across large volumes of data.

- CPUs, GPUs, ASICs, and other specialised architectures each offer different trade-offs between flexibility, throughput, power efficiency, and development time.
- CPUs are well suited to control logic and sequential tasks, while GPUs are attractive for highly data-parallel workloads and ASICs offer high efficiency at the cost of reduced programmability.
- For UAV radar applications, software-defined GPU-based systems are often preferred because they allow algorithms to be updated more easily than fixed-function hardware.
- GPUs achieve high performance by executing many threads in parallel, making them well suited to radar operations with regular memory access patterns and repeated arithmetic across large arrays.
- Their performance depends not only on raw parallelism, but also on architectural factors such as cache behaviour, memory access patterns, branch divergence, and the choice of warp and block sizes.
- In particular, effective use of L2 cache and predictable control flow can significantly improve throughput, while poor memory locality or divergent branches can reduce performance.
- This architectural understanding helps inform the design of our experiments by motivating attention to data layout, kernel structure, and the balance between computation and data movement on embedded GPU platforms.

CHAPTER 3

Literature Review

3.1 UAV Radar System Feasibility

TODO: Develop these to more completely engage with what they're doing. Read the works fully

Sysak et al. [8] present a lightweight X-band monopulse radar designed explicitly for mid-air collision avoidance on medium UAV platforms, addressing the range shortfall that often limits compact frequency modulated continuous wave (FMCW)-based solutions. Their system integrates a multi-channel, forward-looking antenna with an FPGA-based pulse compression on a single 60 gram PCB, enabling online range processing and adaptive CA-CFAR detection, while keeping average power consumption below 80 Watts. Field trials in urban environments demonstrate reliable detection of large obstacles at 1–3 km, providing manoeuvring standoff times on the order of tens of seconds. This work is strongly aligned with UAV Detect-and-Avoid motivations and demonstrates that high-performance sensing can be achieved under strict SWaP constraints with a pulsed, multi-channel architecture.

However, the evaluation is primarily conducted against large, high radar cross-section infrastructure targets, and does not characterise sensitivity to small airborne targets in realistic encounter geometries. In addition, while the work acts as a proof of concept for UAV radar systems, it does not address multi-target tracking nor the latency, throughput and memory usage of a complete onboard pipeline. These omissions are important for DAA, where hardware is limited, and efficient data handling is as critical as raw detection range. The proposed

project complements Sysak et al. by benchmarking detection and tracking sensitivity on small moving targets, and by investigating GPU-oriented implementations and pipeline architectures suitable for low-compute embedded platforms, including the performance impact of memory movement.

3.1.1 End-to-end UAV DAA onboard processing pipelines

Feasibility studies with more emphasis on the radar DAA processing pipelines have been documented. Three that are especially notable include the systems designed by Shi et al. [9], Moses et al. [10], and Newmeyer et al. [11]. Shi et al, similarly motivated by the need to detect-and-avoid non-cooperative targets, describe the design of a small FMCW phased array built mainly from off-the-shelf development boards. The design is only suitable for large UAVs proposed project will target smaller platforms. Additionally, the full radar processing was not entirely onboard and online, splitting steps between an onboard FPGA and offboard post processing. On the other hand, the proposed research will design the software for fully onboard, online tracking.

Moses et al. illustrate the capabilities of a lightweight, single channel FMCW radar. Testing was performed on a quadcopter. A major component of their analysis emphasises identification via micro-Doppler modulation, which is largely evaluated through Doppler-domain signatures over end-to-end range–Doppler detection and tracking. While this is well motivated for classification, it doesn't address the tracking targets in the range–Doppler space. It also remains unclear how the small aperture and SWaP limits affect range-domain SNR, achievable range resolution, and the resulting false alarm behaviour. The proposed research fills this gap by benchmarking range–Doppler processing, CFAR detection, monopulse angle estimation, and multi-target tracking in low-SNR regimes, and by quantifying how these choices trade sensitivity against latency and throughput on embedded GPU hardware.

Finally, Newmeyer et al. demonstrated a 120 gram, 8 Watt FMCW phased array radar and processing pipeline that utilised the FPGA fabric on a Zynq ARM CPU. Power, compute cost, latency distributions, and hardware resources were closely monitored, and the project clearly

demonstrates that radar DAA systems can operate under strict SWaP constraints. However, the study only evaluates one hardware configuration, and one radar algorithm sequence. The proposed research fills this gap by systematically comparing alternative algorithmic pathways and low-level implementations, and by evaluating how pipeline bottlenecks and end-to-end performance change across different embedded compute and memory configurations, including GPU-based architectures.

While these works demonstrate early feasibility of small-platform DAA radar, their hardware stacks are not fully representative of modern embedded radar processing platforms in the sense that they do not utilise GPUs or occur entirely onboard, and their evaluations do not explore how algorithmic selection relates to sensitivity to faint target signals. Consequently, there remains a gap in the literature regarding how to integrate and benchmark alternative radar processing and tracking algorithms for small conformal apertures under SWaP constraints, particularly when optimising jointly for sensitivity, latency, and throughput.

3.2 GPU Radar Processing Performance

TODO: Maybe write about other papers that broadly explore GPU acceleration in the context of any sensors or in the UAV context

3.2.1 GPU-accelerated radar signal processing

TODO: Develop this futher

Studies that are less UAV focused, such as the GPU-accelerated passive bistatic radar pipeline of Zhao et al. [12] and the real-time GPU radar processing thesis of Zaknoun [13] provide more comprehensive analyses of radar software architecture and GPU implementation considerations. Zhao et al. implement compute unified device architecture (CUDA) acceleration

for key stages, reporting up to a 113.13× speedup over a CPU baseline. The research is limited to only three algorithms (ECA-B clutter suppression, range–Doppler processing, and 2D CA-CFAR detection), leaving gaps for similar research that applies to other common radar algorithms such as angle sensing, coordinate transforms, kalman filters, and more. Additionally, the research did not explicitly discuss the time cost of memory reallocation and thread synchronisation. The proposed project will fill these gaps from this study by benchmarking a more comprehensive selection of tracking algorithms for CUDA implementations, and profiling bottlenecks such as host–device data transfer and branching delay to provide a clearer understanding of the benefits of parallelised processing.

3.3 Low SNR Mono-pulse Feasibility

TODO: This is an opportunity to bring in really established papers on monopulse radar and explain what they do to maximise SNR. It may be hard to find gaps that aren't obvious.

TODO: Develop the below further

In addition to a lower SNR, a key limitation of SWaP-constrained monopulse radars is that a reduced aperture degrades angular resolution, increasing the likelihood that multiple targets occupy the same angular cell and that angle estimates begin reflecting the centroid of the target group. An et al. [14] propose an efficient method for resolving two unresolved targets in a monopulse cell, but evaluation is limited to simulated white-noise conditions and does not address integration into an end-to-end range–Doppler detection. The method also has explicit failure modes (same azimuth/elevation; three or more targets in-cell). To address these gaps, our research will test this algorithm with real simulation data, where noise is not necessarily white, and compare how the incorporation of the algorithm into a tracking system improves track quality.

Complementing this, Aubry et al. [15] demonstrate that detection and angle estimation can be treated jointly in a multichannel array, rather than as a strict two-stage pipeline (detect then estimate), which is attractive under SWaP conditions where false alarms and low SNR make early thresholding brittle. The research demonstrates an algorithm that fuses data between channels before angle tracking occurs, increasing sensitivity to targets that are not visible in any single channel. The research exclusively uses single simulated targets in Gaussian interference to validate the algorithm, leaving a gap in characterising the performance under real data. The proposed project will address these gaps by applying the algorithm to alternate noise regimes and multiple target scenarios.

3.4 Programmable Vision Acceleration for Radar

TODO: Find works that explore NVIDIA's PVA in the context of Radar

3.5 Summary

3.5.0.1 End-to-end systems level comparisons under SWaP constraints

Across the UAV DAA radar literature, feasibility has been demonstrated, but evidence remains fragmented across hardware prototypes, single-path processing demonstrations, and isolated GPU acceleration studies. As a result, there remains a gap in consolidated, systems-level research that illustrates which combinations of radar signal processing, monopulse angle estimation, and multi-target tracking algorithms are most effective when jointly optimised for sensitivity, latency, and throughput on GPU-based embedded platforms, and how these conclusions change with hardware configuration. The proposed evaluation of small-target sensitivity and end-to-end tracking on embedded GPU hardware directly supports our contributions on algorithm selection, low-level implementation trade-offs, and hardware-aware design recommendations.

3.5.0.2 Realistic small-target and encounter-focused evaluation

The experimental design explicitly includes small moving aerial targets and encounter-like scenarios in both simulation and real data collection. Performance is measured not only by detection range but by false alarm behaviour, track continuity, and sensitivity in low-SNR regimes that better reflect small-aperture UAV radars. By extending prior single-path demonstrations into a comparative study of alternative algorithm combinations and fully onboard GPU implementations, this section motivates our contributions on algorithm selection and implementation choices, and on how these choices vary across hardware contexts.

3.5.0.3 GPU-oriented onboard pipeline evidence, including data movement costs

The thesis targets embedded GPU hardware and evaluates architectural choices that often dominate real-time performance, including host–device transfers, memory layout, streaming formats, kernel fusion opportunities, and synchronisation overhead. This thesis advances the literature by benchmarking low-level GPU implementations beyond range–Doppler and CFAR, quantifying end-to-end latency and throughput impacts, and assessing how these conclusions change across embedded hardware configurations.

3.5.0.4 Hardware-aware design conclusions across embedded configurations

Experiments are repeated across multiple hardware contexts (for example CPU-only, unified-memory embedded GPU, and discrete CPU+GPU settings where available) and across constrained power and clock modes. Testing these angle-estimation methods within a complete multi-target tracking pipeline, and measuring both sensitivity and track-quality impacts alongside compute cost, directly supports our contributions on algorithm selection, implementation trade-offs, and hardware-aware recommendations.

CHAPTER 4

Methodology

- The methodology of this thesis involves designing and implementing a software-defined radar processing pipeline for a small monopulse radar system, and evaluating its performance in terms of both computational efficiency and tracking accuracy.
- This section firstly outlines all the difference techniques we will use to evaluate the performance of our software, including the datasets we will use for testing, the metrics we will report, and the tools we will use for performance analysis.
- We then outline what algorithms we are applying these validation techniques to and how these experiments fulfil the goals of the thesis.

4.1 Testing Datasets

Three types of datasets are used for the experiments outlined in the following sections (see Sections 4.5, 4.6, and 4.7):

- **Generic Sine Waves with Noise**: To validate the correctness of our processing algorithms as well as their speed, we generate binary files of synthetic radar data consisting of a complex sine wave with additive white Gaussian noise. This allows us to test the software under controlled conditions where we can precompute the expected output, and to evaluate the performance of the software in terms of latency and throughput without the confounding factors of real-world data. (INCLUDE FIGURE HERE illustrating the signal)
- **Simulated Radar Data**: To evaluate the performance of our algorithms under more realistic conditions with a known ground truth, we use simulated radar data

generated by the Ocean Acoustics library. This library allows us to model the radar signal propagation, target reflections, and noise characteristics in a way that closely mimics real sonar and radar returns. By using this dataset, the accuracy of stages such as monopulse angle discrimination can be evaluated with realistic phases and amplitude differences between channels. Furthermore, this library allows us to simulate multiple targets and clutter, allowing for full integration tests and visualisations of the tracking performance in real time.

- **Real Radar Data**: To validate the performance of our software in real-world conditions, we use data collected from AWR radar chips. This dataset includes both false target data, where the chip outputs a clean compressed pulse corresponding to an imaginary target, and real target data (where actual targets are present). Testing with this dataset allows us to evaluate the robustness of our algorithms to real-world (non-Gaussian) noise, unique target signatures, and realistic clutter, as well as, potential timing, data corruption and memory issues that may arise when running the software in real time.

4.2 Performance Analysis

4.2.1 Throughput Analysis

TODO: Measure in Bytes Per Second

4.2.2 Latency Analysis

TODO: Use NSight

TODO: Ablation Study

TODO: Explore the effect of different hardware contexts on the performance. What contexts prevent us from implementing certain solutions and why

4.3 Performance Analysis

- This section evaluates the computational performance of the radar pipeline, focusing on throughput and latency.
- The aim is to determine whether the implemented algorithms are suitable for real-time UAV DAA applications.
- Throughput analysis measures how much data each implementation can process per unit time, reported in bytes per second.
- This shows whether the pipeline can sustain the required radar data rate.
- Latency analysis measures the time taken for data to pass through each stage of the pipeline and through the system as a whole.
- This is profiled using tools such as NVIDIA Nsight to identify bottlenecks, memory transfer costs, and kernel execution overheads.
- Variations of each operation are compared to understand the impact of different algorithmic choices and optimisations on performance.
- An ablation study is used to evaluate the impact of individual optimisation choices by noting how the performance of the pipelines change when specific operations are removed from the process.
- This helps identify which design decisions provide meaningful performance gains, and where further optimisation efforts should be focused. A table will illustrate the latency and throughput contribution of each stage of the pipeline.
- Performance is also compared across different hardware contexts to understand how architectural constraints affect implementation choices and runtime behaviour, e.g. physically unified memory vs managed memory.
- This informs which solutions are practical on embedded and GPU-based radar platforms.

4.4 Tracking Performance Analysis

- This section evaluates how different algorithmic choices affect tracking performance, not just computational speed.
- In addition to runtime analysis, we compare the effect of each design on sensitivity in detection and tracking tasks.
- An ablation study is used to test alternative implementations for key operations and measure their impact on tracking quality.
- One framework that allows the testing of many different tracking algorithms is Stone Soup. The tracking stages of the pipeline are implemented in Stone Soup, which allows us to test for correctness and performance across a variety of tracking algorithms.
- This allows us to identify which optimisations preserve accuracy and which introduce unacceptable performance loss.
- After both computational and sensitivity analysis, a trade-off table is produced to compare candidate pipeline designs.
- This table summarises the computational cost and tracking performance of each design and is used to inform the final embedded implementation.

4.5 Processing Application

TODO: Complete the following sections. How do the experiments resolve the gaps in the literature.

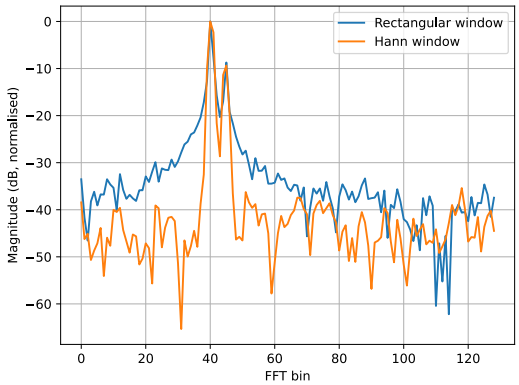

FIGURE 4.1: Results of an FFT performed on a signal with two frequencies and Gaussian noise with and without a Hann window. Without the Hann window there bins that are close to the peak frequency bins 40 and 45 are significantly larger than the bins further away, which misrepresents the homogeneity of the Gaussian noise

4.5.1 Data Input

4.5.2 Shaping Data

4.5.3 Windowing

4.5.4 Fast Fourier Transforms

4.5.5 Coherent Integration

4.5.5.1 Windowing

To reveal target range from a signal radar pulse, we need to take an FFT (Fast Fourier Transform) of the dechirped pulse. When an FFT is performed on a finite signal, the theoretically infinite signal is implicitly multiplied by a rectangular window. A term-wise multiplication in time is a convolution in the frequency domain. A rectangular window has large spectral sidelobes, resulting in a smearing of the original signal's frequencies. By windowing our signal more strategically, we can minimise the smearing of frequencies (see Figure 4.1).

This thesis will evaluate a range of practical window functions and their implementation on CPU and GPU. Candidate windows include common real-time tapers such as Hann and Hamming, lower-sidelobe windows such as Blackman and Blackman–Harris, and adjustable-parameter windows (e.g., Kaiser) that allow explicit control of sidelobe level versus mainlobe width. Because windowing is a point-wise multiplication, it is naturally suited to single instruction, multiple data (SIMD) execution.

Performance profiling will measure the latency contribution and throughput associated with each filter and implementation method. A cost benefit analysis will weight the latency cost against the downstream benefits of a cleaner signal, such as less sidelobe-induced false positives. These procedures will provide a basis for selecting windows that are spectrally well behaved, computationally efficient, reduce detection false alarms, and reduce the computation workload of downstream algorithms such as CFAR.

4.5.5.2 FFTs

FFTs are an essential processing step in any radar processing pipeline, as they convert time series signals to target ranges. The interplay between the asymptotic performance of the FFT algorithm as an $\mathcal{O}(N \log N)$ operation, and the constant and linear time memory access patterns on CPU and GPU implementations make the optimal FFT length, batching strategy, and dimensionality unclear. While [6] assessed an implementation that used an dedicated GPU, GPU with shared memory, and more modern NVIDIA accelerators such as the programmable vision accelerator may show differ performances, that the proposed project will measure. This thesis will compare CPU and GPU FFT implementations under streaming workloads using optimised libraries (e.g., FFTW/MKL on CPU and cuFFT on GPU with shared memory, and using the PVA). Trade-offs will be analysed by relating FFT choices to latency, throughput, memory usage, and downstream detection quality.

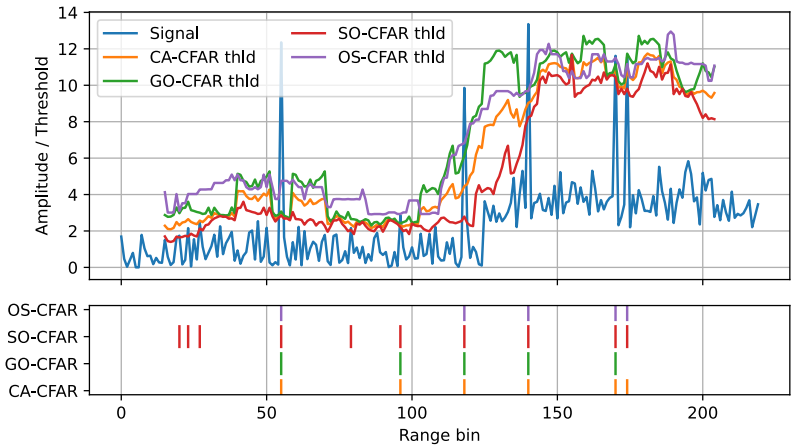

FIGURE 4.2: Performance of four CFAR variants. Each algorithm registers a detection when its threshold is below the signal. The test signal has 5 peaks that are each surrounded by different noisy distributions. The bottom plot illustrates the detections made by each algorithm. OS-CFAR performs well in this example.

4.5.5.3 CFAR

Constant False Alarm Rate (CFAR) detection is a key mechanism for resolving a map of noisy range data into discrete detections given a target false alarm rate. It does this by masking out any samples with magnitude below a threshold determined by the power of the samples around it. There are several variants of CFAR that each require consideration (see Figure 4.2). Cell averaging (CA)-CFAR is the most appropriate for homogeneous sample data, greatest of (GO)-CFAR performs well with cluttered edges, smallest of (SO)-CFAR is suitable for multi target tracking, and ordered statistic (OS)-CFAR is appropriate for non-homogenous multi-target settings. A sweep of window size and false alarm rate values will be conducted for each CFAR mode on real and simulated data and on the CPU and GPU. The performance (throughput, latency, memory, false alarm rate, sensitivity) will be compared across configurations in a cost benefit analysis. This addresses the literature gap in understanding how detection sensitivity and false alarm behaviour trade off against latency and throughput under SWaP constraints.

4.5.5.4 Time Integration

Time integration accumulates energy across pulses to improve effective SNR for detection and tracking. We will compare coherent and non-coherent integration on real data, measuring how detection sensitivity and false alarm behaviour change as integration increases. We will compare CPU and GPU implementations, and quantify the downstream impact on angle sensing stability. An ablation study will hold the window, FFT size, and CFAR parameters fixed while varying only the integration policy to isolate its effect. Different locations for the integration to take place in will also be compared.

4.6 Tracking Application

TODO: Complete the following sections. How do the experiments resolve the gaps in the literature.

4.6.1 CFAR

Constant False Alarm Rate (CFAR) detection is a key mechanism for resolving a map of noisy range data into discrete detections given a target false alarm rate. It does this by masking out any samples with magnitude below a threshold determined by the power of the samples around it. There are several variants of CFAR that each require consideration (see Figure 4.2). Cell averaging (CA)-CFAR is the most appropriate for homogeneous sample data, greatest of (GO)-CFAR performs well with cluttered edges, smallest of (SO)-CFAR is suitable for multi target tracking, and ordered statistic (OS)-CFAR is appropriate for non-homogenous multi-target settings [16]. A sweep of window size and false alarm rate values will be conducted for each CFAR mode on real and simulated data and on the CPU and GPU. The performance (throughput, latency, memory, false alarm rate, sensitivity) will be compared across configurations in a cost benefit analysis. This addresses the literature gap in

understanding how detection sensitivity and false alarm behaviour trade off against latency and throughput under SWaP constraints.

4.6.2 Monopulse Angle Sensing

Angle sensing converts multi-channel measurements into angle estimates that drive association and tracking. We will assess candidate angle discriminators by running controlled sweeps over SNR, channel imbalance, and multi-target proximity, and measuring angular bias, mean squared error, and confidence. The latency and throughput of CPU and GPU implementations will be measured. We will explicitly quantify the time and compute costs of stabilisation strategies such as sum-channel gating, regularised ratios, and calibration placement. The "Amplitude Angle Discriminator" and "Phase Angle Discriminator" methods will be compared for their computation requirements, angular bias and variance [17]. The algorithm will be validated on data obtained from flight tests. This addresses the gap in validating monopulse angle estimation methods beyond idealised simulations, which often assume Gaussian noise and clean single detections per target.

4.6.3 Polar Transform

Because the radar have multiple receiver channels, each target detection will need to be rotated from the receiver's local polar coordinate from to the radar module's polar coordinate frame. We will also compare alternative polar-to-Cartesian transformation strategies in terms of numerical stability, computational cost, and their effect on downstream tracking accuracy. This addresses the gap in understanding how coordinate conversion choices influence both runtime performance and state estimation quality in real-time radar pipelines.

4.6.4 Tracking Filter

Filter choice determines how measurements are fused over time and how manoeuvres are handled. We will assess candidate filters by comparing extended Kalman filters (EKF) [18], unscented Kalman filters (UKF) [19], and interacting multiple models (IMM) [20]

filters implemented on the CPU and GPU. Our evaluation metrics will include mean squared error (MSE), normalised estimation error squared (NEES), normalized innovation squared (NIS), latency, average time-to-confirm tracks, fragmentation [21], and ID-switch rate under clutter, missed detections, and unexpected manoeuvres. The evaluation will report how filter complexity translates into measurable improvements in track quality, and whether those gains justify the added costs.

4.6.5 Data Association

Data association assigns detections to existing tracks in multi-target scenarios. We will compare nearest neighbour (NN), global nearest neighbour (GNN) assignment, and joint probabilistic data association (JPDA) under sweeps of target density and clutter. Performance will be assessed using the "CLEAR MOT" metrics for multi-target tracking algorithms [22] alongside computational metrics.

4.6.6 Embedding

The final radar pipeline must operate within the memory, power, and latency constraints of embedded hardware. We will evaluate how candidate pipeline designs transfer from desktop development platforms to embedded targets, measuring throughput, latency, memory use, and any loss in detection or tracking performance. This addresses the gap between algorithm validation in high-resource environments and practical deployment under SWaP-constrained UAV settings.

4.6.7 Initiating and Deleting

Track initiation and deletion logic strongly affects tracker stability, especially in cluttered scenes with missed detections and short-lived false alarms. We will compare alternative confirmation and deletion rules using metrics such as time-to-confirm, false track rate, fragmentation, and track persistence. This addresses the gap in understanding how track life-cycle design influences the balance between responsiveness and robustness in UAV radar tracking.

4.7 Orchestration and Dataflow

- This section examines how radar data is moved, buffered, and scheduled through the pipeline, and how these choices affect latency and throughput.
- **Threads and Batching:** We will compare threading and batching strategies to understand the trade-off between throughput and end-to-end delay.
- **Reading data:** As a baseline, input will be read from `stdin` streams, then compared against lower-overhead approaches such as direct memory access into kernel buffers. The `vivid` virtual video device will be used for controlled testing.
- **Writing data:** We will compare whether stages should communicate through standard application pipes or through lower-copy shared-memory style interfaces.
- **Aggregating Data Streams:** Data from the four AWR chips must be combined with minimal delay while remaining robust to missing or late packets. We will compare strict synchronisation against timeout-based and best-effort aggregation policies.

4.8 Results

TODO: Mirror the methodology

4.9 Processing Application

4.9.1 Data Input

4.9.2 Shaping Data

4.9.3 Windowing

4.9.4 Fast Fourier Transforms

4.9.5 Coherent Integration

4.10 Tracking Application

4.10.1 Constant False Alarm Rate

4.10.2 Monopulse Angle Sensing

4.10.3 Embedding

4.10.4 Polar Transform

4.10.5 Data Association

4.10.6 Tracking Filters

4.10.7 Initiating and Deleting

4.11 Orchestration and Dataflow

4.11.1 Threads and Batching

4.11.2 Reading data

4.11.3 Writing data

CHAPTER 5

Discussion

5.1 Orchestration and Dataflow

TODO: Identify the best strategy for each of the subsections. In terms of our different performance metrics.

TODO: What do our tools and result suggest is happening on a low level. Come up with explanations for any phenomina which isn't immediately obvious (i.e. most things)

TODO: How do these findings relate to the results to the other sections. Can we fuse any operations together or take any cleaver short cuts

TODO: What hardware allows us to do which options, i.e. Direct Memory Access, Shared Memory

TODO: Explicitely address the knowledge gaps and contributions

5.1.1 Threads and Batching

5.1.2 Reading data

5.1.3 Writing data

5.1.4 Aggregating Data Streams

5.2 Processing Application

TODO: Discuss the results for each of the operations. relate to our contributions, gaps and results from the other areas of research.

5.3 Tracking Application

TODO: Discuss the results for each of those. relate to our contributions, gaps and results from the other areas of research.

5.4 Real World Implications

TODO: Discuss how our findings either support or don't support the feasibility of radar systems for small UAVs. What size UAVs does this work for? How should policy be written based on our findings. How should people feel about the safety.

TODO: Cultural, Social, Policy, Environmental Implications. Ethics

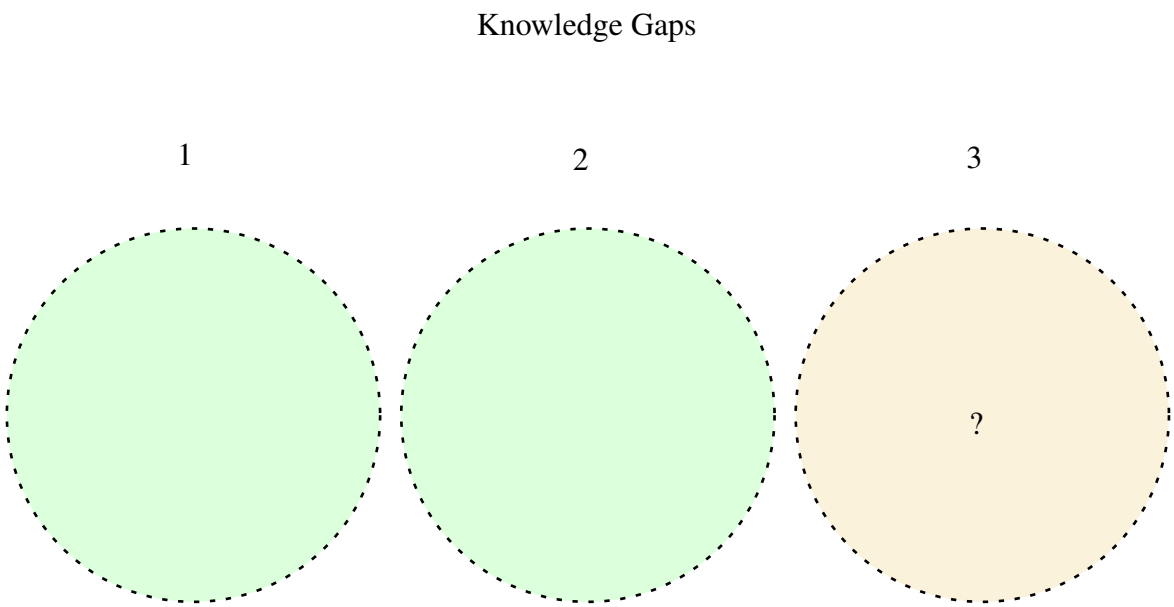

FIGURE 5.1: Me just messing around with Inkscape

TODO: Recommendations for Engineers.

5.5 Addressing the Identified Knowledge Gaps

TODO: Explicitely address the knowledge gaps

CHAPTER 6

Conclusion

6.1 Summary of Findings

TODO: Summarise the finds at a high level. State the best/recommended configuration.

TODO: Link together all the sections of the discussion into main takaways. Kinda like an executive summary.

6.2 Industry Partner Outcomes

TODO: Without sharing anything sensitive, detail how the research done benefits mission systems in their project goals.

TODO: Document tests and downstream work that my work enabled.

TODO: Discuss how this technology will after the industry outside of Mission Systems too.

6.3 Future Work

TODO: What about our research was successful, what wasn't successful

TODO: Discuss the limitations of the set of computing hardware that we were able to explore

TODO: We only explored one radar device

TODO: Talk about algorithms that were not explored

TODO: Compare to other sensing systems

TODO: Sensor Fusion

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APPENDIX A

Appendix

A1 Case Studies

A2 Case Studies: Learning Objectives Covered by Thesis

For ESIPS substitution, this thesis is proposed to replace content from two elective units: **AMME5520 Advanced Control and Optimisation** and **ELEC3305 Digital Signal Processing**. The proposed thesis covers over 75% of learning objectives from each unit outline through the necessary self-guided learning and engineering design required to meet the contributions.

A2.1 Case Study 1: AMME5520 Advanced Control and Optimisation

A2.1.1 LO1. *Approach research papers in a professional and research-orientated manner, and conduct critical reviews of these papers*

This learning objective is met through the thesis literature review and gap synthesis across UAV DAA radar pipelines, GPU radar processing, monopulse angle estimation under low SNR, and multi-target tracking. The thesis requires evaluating assumptions, identifying limitations, and justifying design choices based on evidence, aligning with Week 13 Lightning talks (LO1) and the unit's self-directed critical engagement with research.

A2.1.2 LO2. *Implement simple path generation algorithms, controllers and decision metrics for an autonomous system in order to meet specific mission objectives*

This learning objective is met by implementing autonomy-relevant estimation and decision components in the detection-to-track chain: measurement formation, data association (e.g., NN/GNN/JPDA variants), tracking filters (EKF/UKF/IMM-style variants), and track lifetime management. Although the thesis project does not explicitly control the position of the targets that it is making estimations on, estimation is an essential component of any control system, and multi target tracking engages with the state based decision making that this learning objective is concerned with. The estimation theory in this project has heavy alignment with Weeks 07–08 in particular (Kalman filtering and extensions).

A2.1.3 LO3. *Understand a number of different path generation and control algorithms implemented in autonomous systems and how they are linked to optimality criteria, platform stability and vehicle constraints.*

Tracking aerial vehicles that can both fly in a straight line and conduct several manoeuvres motivates the use of multiple filters and tracks that model possible trajectories of the targets. Although targets are not being controlled, multi target tracking itself is an optimisation problem. Understanding stability is essential to minimise track id switching when multiple targets are within a few standard deviations of one another (in terms of state).

This learning objective is also met through the comparative evaluation of tracking algorithms, where we optimise for high track quality and low computational resource use. This aligns with Weeks 02–03 (optimality-driven algorithms), Weeks 07–09 (estimation, uncertainty, robustness).

A2.1.4 LO4. *Understand how cost functions are used to define mission objectives in a mathematical form, so that autonomous systems can make decisions about their next action.*

This learning objective is met by framing pipeline design as an objective-driven trade-off between sensitivity/track quality and computational cost, using ablations and parameter sweeps to identify design points. This aligns with Week 01 (optimal control framing), Weeks 10–11 (optimisation/MPC), and the unit's focus on formal objective definition. Additionally, track association is an optimisation problem.

A2.1.5 Coverage statement

The thesis addresses LO1–LO4 through critical review, implementation of estimation/decision algorithms, comparative evaluation under constraints, and objective-driven trade-off analysis, demonstrating at least 75% coverage.

A2.2 Case Study 2: ELEC3305 Digital Signal Processing

A2.2.1 LO1. *Describe signals using mathematical concepts such as convolutions, transforms, spectral analyses, linear difference equations, filters, correlation and covariance*

This learning objective is addressed through frequency-domain radar processing. Convolutions will be used in the form of matched filters in the dechirping process. The Fourier transform will be applied with the FFT algorithm over both fast time and slow time to produce range and Doppler maps. Hann, Hamming, and other filters will be applied to minimise spectral leakage, and the spectral effect of our hardware configuration and geometry will be compensated for. Correlation, linear difference, and covariance will be essential concepts in the process of detection and tracking of targets. This aligns strongly with Week 03 (convolution and frequency response), Weeks 07–10 (DFT/FFT properties and spectral analysis), and the unit's emphasis on transform-based signal description.

A2.2.2 LO2. *Use software, such as MATLAB and Python, to develop code to analyse and process signals including filtering, inverse filtering, spectral analyses, resampling, time-frequency analyses*

This learning objective is addressed through substantial software implementation and testing of DSP stages in an end-to-end pipeline, including benchmarking CPU vs GPU implementations and developing validation/profiling tools with parameter sweeps. This aligns with Weeks 02–12 practicals, which focus on implementing DSP in software. Python and C++ will be used to develop both the pipeline and the simulation application.

A2.2.3 LO3. *Identify linear signal models for time-invariant systems such as convolutional models, sinusoidal and harmonic models, echo and reverberation models, modulation models and time-frequency models.*

This learning objective is addressed by using standard linear and time-frequency models for radar returns and mapping sampled signals into range–Doppler representations, then studying how model behaviour changes under low SNR and limited resolution. This aligns with Week 02 (signal models), Weeks 04–06 (DTFT/Z-transform/transfer functions), and Weeks 11–12 (sampling and ADC/DAC considerations).

A2.2.4 LO4. *Design and evaluate signal processing systems.*

This learning objective is addressed directly by designing and evaluating alternative pipeline configurations (windowing, FFT sizing/batching, CFAR variants, integration policies, and angle sensing), and quantifying effects on detection sensitivity, false alarms, and end-to-end latency/throughput. This aligns with Weeks 04–10 and the unit's system design emphasis.

A2.2.5 Coverage statement

The thesis covers LO1–LO4 through transform-based processing, software implementation, linear/time-frequency modelling, and end-to-end system design and evaluation, demonstrating at least 75% coverage.

A3 Work Health and Safety Report

TODO: Write the health and safety report

TODO: I don't quite know what this involves, but it is worth 5% so I shouldn't ignore it

A3.1 Work, Health and Safety at Mission Systems

A3.2 Incidents and Preventions During Placement

A4 Practical Experience Logbook